



Roaming for Applications

M1

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CONTEXT AND PROBLEM

Roaming



Edge Computing



European Initiatives



seba é gay

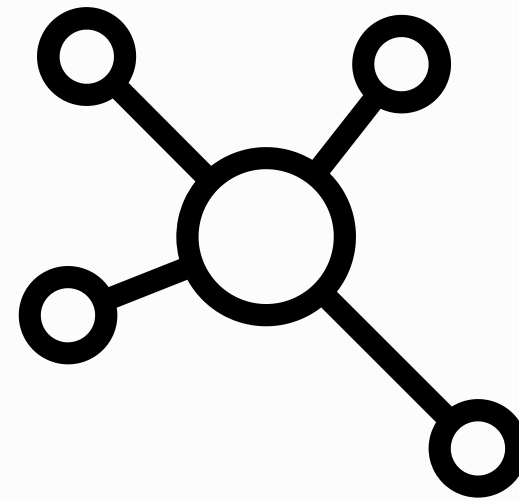
GOALS AND EXPECTED RESULTS

REQUEST



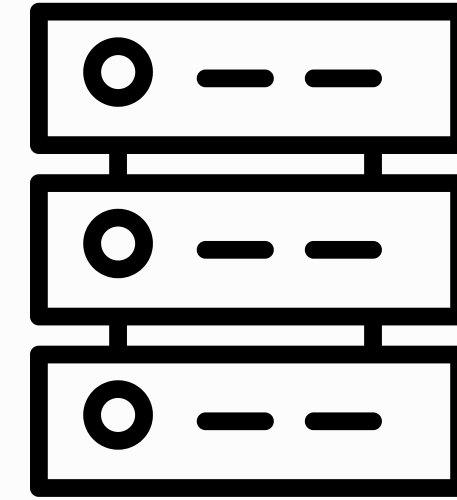
**Request through
one interface**

DELEGATE



**Automatic Partner
Selection**

RUN



**Extend Application
Coverage**

**Requested through Operator A. Running on Operator B.
Managed from one interface.**

Task List

Research (Everyone)

Task 1 (Everyone): Read IPCEI-CIS Reference Architecture (ICRA)

Task 2 (Everyone): Read GSMA v7.0 with special focus on Federation and E/WB Interfaces

Task 3 (Margarida Pinheiro): Related Work (State of the Art Draft)

Task 4 (Margarida Pinheiro): More Detailed State of the Art

Implementation Modules

Task List

Platform Infrastructure and Operator Simulation (Miguel Sousa)

Task 1 (Miguel Sousa): Request IT cloud access and resources

Task 2 (Alan Marques & Miguel Sousa): Set up Kubernetes clusters for 3 operators

Task 3 (Miguel Sousa): Write Dockerfiles for the API, FM and Deployment Layer

Task 4 (Miguel Sousa): Define simulated operators: name, locations, capacity, latency values

Federation Manager (Margarida Pinheiro)

Task 1 (Margarida Pinheiro): Define the FM's internal architecture and its interfaces to the API and Deployment Layer

Task 2 (Margarida Pinheiro & Alan Marques): Build the Deployment Layer mock so FM implementation progress can evolve independently

Task 3 (Margarida Pinheiro): Federation Catalog: data model and local publishing of the operator's offering

Task 4 (Margarida Pinheiro & Rodrigo Sampaio): Hand-over decision logic: local if capacity fits, otherwise pick a partner from its catalog

Task 5 (Margarida Pinheiro & João Ribeiro): Track application state and propagate status back to the API

Task 6 (Margarida Pinheiro): Develop tests along the way

East/West Bound Interface Implementation (Rodrigo Sampaio)

Task 1 (Rodrigo Sampaio): Implement the server side (endpoints exposed to partners)

Task 2 (Rodrigo Sampaio & João Ribeiro): Implement the client side (calls to partner FMs)

Task 3 (Rodrigo Sampaio): Federation onboarding messages

Task 4 (Rodrigo Sampaio & Miguel Sousa): Catalog exchange

Task 5 (Rodrigo Sampaio & Miguel Sousa): Hand-over request, status update and termination messages

Task 6 (Rodrigo Sampaio): Error handling: partner down, rejection, timeouts

Task 7 (Rodrigo Sampaio): Develop tests along the way

Implementation Modules

Task List

Developer-Facing API (João Ribeiro)

Task 1 (João Ribeiro & Alan Marques): Define the application descriptor (image, ports, resources, location, latency), shared with the Deployment Layer

Task 2 (João Ribeiro): Implement the endpoints: register, deploy, status, delete

Task 3 (João Ribeiro): Request validation and error responses

Task 4 (João Ribeiro & Miguel Sousa): Connect it to the FM

Task 5 (João Ribeiro): Generate and publish OpenAPI docs

Task 6 (João Ribeiro): Develop tests along the way

Application Deployment Layer (Alan Marques)

Task 1 (Alan Marques & Margarida Pinheiro): Integrate the internal interface with the FM with endpoints like deploy, status, delete, get_capacity

Task 2 (Alan Marques): Implement container deployment with Kubernetes (consider VM integration later)

Task 3 (Alan Marques): Implement lifecycle and status reporting

Task 4 (Alan Marques): Develop tests along the way

Project Specification Modules

Task List

○ Architecture (Miguel Sousa)

Task 1 (Miguel Sousa): Define the modules and their interactions

Task 2 (Miguel Sousa): Decide shared representations like location, capacity, application descriptor

○ East/West Bound Interface Specification (Rodrigo Sampaio)

Task 1 (Rodrigo Sampaio): Define the list of messages and their purpose

Task 2 (Rodrigo Sampaio): OpenAPI/Swagger Specification

Task 3 (Rodrigo Sampaio): Document the flows: onboarding, catalog exchange, hand-over, status

○ Developer-Facing API Specification (João Ribeiro)

Task 1 (João Ribeiro): Document endpoints and the application descriptor

Task 2 (João Ribeiro): Add request and response examples

Project Specification Modules

Task List

Technical Report (Everyone)

Task 1 (Everyone): Agree on the outline and shared template

Task 2 (Alan Marques): Draft the introduction, context and related work

Task 3 (Everyone): Everyone keeps updating the report as we do our respective tasks

Task 4 (Everyone): Final review

Project Administration Modules

Task List

○ Project Management and Communication (Rodrigo Sampaio & Margarida Pinheiro)

Task 1 (Margarida Pinheiro): Set up the repository and give supervisors access

Task 2 (Margarida Pinheiro): Set up the project website

Task 3 (Rodrigo Sampaio): Run regular team meetings, keep notes and manage tasks

○ Presentations (Everyone)

Task 1 (Margarida Pinheiro): Draft presentation template

Task 2 (Everyone): Everyone adds in the presentation their respective work for each respective milestone

○ Demo (Alan Marques)

Task 1 (Alan Marques): Design the demo scenario

Task 2 (Alan Marques): Dry run on the IT cloud

CALENDAR

Weeks 2–4

Week 2

22/09/2026



Team and project decided. First meeting with supervisors to understand goals and scope of project. Reading of important documentation in relation to the IPCEI-CIS and basic understanding of the current related work.

Week 3

29/09/2026



M1: Definition of context, goals, calendar, repository and tasks for each team member. First presentation done.

Week 4

06/10/2026



Better understanding of the ICRA. Investigate the ETSI SDG Open OP. Definition of requirements and use cases. Project report initiation.

CALENDAR

Weeks 5–7

Week 5

13/10/2026



M2: More precise definition of the architecture, technologies and how the Federation layer will be developed. Second presentation done.

Week 6

20/10/2026



Implementation of the Provider Access Point API and core logic for a single platform instance, with basic application deployment layer, following the architecture and technologies defined in M2.

Week 7

27/10/2026



Implementation of more platform instances, ideally 3, all with basic application deployment layers. Review/change of the calendar and objectives based on current work achieved.

CALENDAR

Weeks 8–11

Week 8 & 9
03/11/2026



MS3: Start of development of the Federation Manager and integration with the deployment layer. Draft of the EWBI API spec. Third presentation done.

Week 10
17/11/2026



Federation Manager core logic developed with discovery and onboarding between the platforms, start of the implementation of the hand-over decision logic to other partner platform instances. Tests development.

Week 11
24/11/2026



Federation Manager logic refined. Hand-over logic triggers the deployment on the partner platform. Updated tests.

CALENDAR

Weeks 12–14

Week 12 & 13

01/12/2026



Hand-over mechanisms working, including application instantiation on the partner and status reported back. Refinement and polishing of the report. Updated tests.

Week 14

15/12/2026



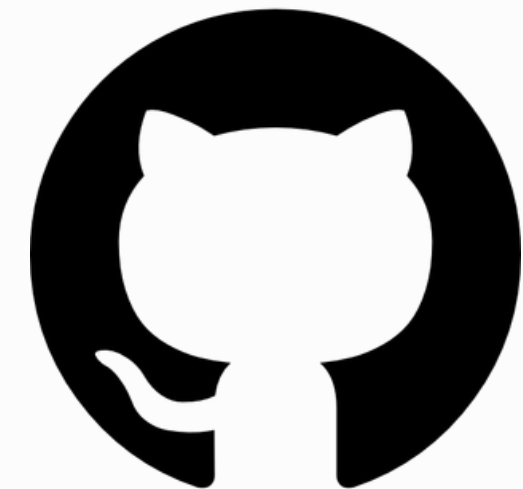
MS4: Review and changes in the project, current code reviewed/polished, and draft project report. Assessment of work achieved. Fourth presentation done.

Related Work

					Our Poject
Inter-Operator Federation					
Latency & Location Aware Decision					
Transparent Application Handover					
Single Developer Entry Point (PAP)					
Executable Prototype					

Communication

Plan



Our Team



Margarida Pinheiro
DevOps



Alan Marques
QA Engineer/Tester



Rodrigo Sampaio
Team Manager



João Ribeiro
Product Owner



Miguel Sousa
Architect

Advisors

Diogo Gomes & Rafael Direito

Thank You